Climate Risk-Averse Power System Planning Under Renewable Drought Events

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***Abstract*—With increasing shares of renewable energy, episodes of continuously low renewable output, referred to as renewable drought (RD) events, pose growing challenges to the long-term adequacy of renewable-dominant power systems. This paper develops a meteorology-driven framework to characterize the spatiotemporal features of RD events and represent the inadequacy risk arising from their multi-day persistence and cross-regional dependence. Based on this, a risk-aware planning model is formulated to integrate typical-day operation with RD events of varying durations in long-term planning. Two case studies on the modified Garver's 6-node system and China's East Power Grid show that increasing renewable penetration amplifies inadequacy risk under RD events, while coordinated flexibility resources—including inter-regional transmission, long-duration storage, and diversified wind–solar portfolios—can effectively mitigate such risk.**

***Index Terms*—Renewable drought, long-term inadequacy risk, long-duration energy storage, power system planning, spatiotemporal distribution.**

NOMENCLATURE

A. Indicies and Sets

s/t	Index of renewable drought scenario/hourly time step.
r	Index of region.
$g/l/b$	Index of generation unit/transmission line/storage.
d/u	Index of natural/typical day.
$\mathcal{G}_n/\mathcal{B}_n$	Set of generation units g/storage b connected at bus n.
$\mathcal{N}_r$	Set of bus node n locating at region r.
$\mathcal{L}_n^+/\mathcal{L}_n^-$	Set of transmission lines l starting from/ending at bus n.

B. Parameters

$c_g^{\text{inv}}/c_l^{\text{inv}}/c_b^{\text{inv}}$	CAPEX of generation unit g/transmission line l/storage b.
c_g^{ope}	Fuel cost of generation unit g.
c_g^{up}	Start-up cost of generation unit g.
$c_n^{\text{cur}}/c_r^{\text{cur}}$	Load curtailment cost at bus n / region r.
$\omega_{r,u,t}/\omega_{r,s,t}^{\text{rd}}$	Renewable output curve within region r during typical day u / RD scenario s at hour t.
$d_{n,u,t}/d_{n,s,t}^{\text{rd}}$	Load curve at bus n during typical day u / RD scenario s at time step t.
$G_g^0/L_l^0/B_b^0$	Existing capacity of generation unit g / transmission line l / storage b.
T_b	Continuous discharging hour of storage b.
$T_g^{\text{on}}/T_g^{\text{off}}$	Minimum on/off hours for thermal unit g.
$\gamma_g^{\text{up}}/\gamma_g^{\text{dn}}$	Ramp up/down ratio of thermal unit g.
$\eta_b^{\text{cha}}/\eta_b^{\text{dis}}$	Charging/discharging efficiency of storage b.
λ_g	Minimum output ratio of thermal unit g.

C. Variables

$G_g/L_l/B_b$	Expansion capacity of generation unit g / transmission line l / storage b.
$p_{g,u,t}/p_{g,s,t}^{\text{rd}}$	Power output of generation unit g during typical day u / RD scenario s at hour t.
$f_{l,u,t}/f_{l,s,t}^{\text{rd}}$	Power flow of transmission line l during typical day u / RD scenario s at hour t.
$p_{b,s,t}^{\text{cha,rd}}/p_{b,s,t}^{\text{dis,rd}}$	Charging/discharging power of storage b during RD scenario s at hour t.
$p_{b,u,t}^{\text{cha}}/p_{b,u,t}^{\text{dis}}$	Charging/discharging power of storage b during typical day u at hour t.
$d_{n,u,t}^{\text{cur}}/d_{n,s,t}^{\text{cur,rd}}$	Load curtailment at bus n during typical day u / RD scenario s at hour t.
$o_{g,u,t}/o_{g,s,t}^{\text{rd}}$	Online capacity of generation unit g during RD scenario s at hour t.
$o_{g,s,t}^{\text{rd,up}}/o_{g,s,t}^{\text{rd,dn}}$	Start-up/shutdown capacity of generation unit g during RD scenario s at hour t.
$o_{g,u,t}^{\text{up}}/o_{g,u,t}^{\text{dn}}$	Start-up/shutdown capacity of generation unit g during typical day u at hour t.
$q_{b,u,t}/q_{b,s,t}^{\text{rd}}$	Inventory of storage b during typical day u / RD scenario s at hour t.

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D. Function

$\Pi(\cdot)$	Clustering function indicating the mapping relationship from natural days to typical days.
$\mathrm{CVaR}_{r,\alpha}(\cdot)$	Calculation function of conditional value at risk for region r at confidence level α.
$\mathrm{ave}(\cdot)$	Averaging function.

I. Introduction

THE rapid expansion of renewable energy brings increasing challenges to the adequacy of power systems [1]. In future renewable-dominant power systems, renewable drought (RD) events [2], [3], characterized by low solar and/or wind production over an extended period, have caused growing concerns throughout the world [4], [5]. In Australia, for example, a wind drought event lasted for nearly seven days in the early summer of 2024 [6]. In China, a low-solar event hit a 15-day record in the southern provincial power grid in 2007 [7]. In the future, renewable-dominant power systems will experience increasing frequency and intensity of RD events, significantly threatening supply adequacy. The RD effect will be exacerbated by the ongoing retirement of more dispatchable thermal plants and a lack of long-term flexibility measures [8]. Therefore, it is necessary to quantify the long-term inadequacy risk of the power system accounting for RD events and investigate potential solutions to mitigate the risk.

Renewable drought events are closely related to regional meteorological phenomena. A primary example is air stagnation, which causes low wind output. Air stagnation indicates meteorological conditions with limited vertical and horizontal air movement, leading to light winds and a lack of precipitation [9]. As for solar, PV outputs are severely diminished under cloud cover or haze, sometimes lasting for several days or weeks. The term *dunkelflaute* is used to refer to situations with simultaneously low wind and solar outputs [5]. Exploring the spatiotemporal distributions of RD events in the power grid is required to quantify risks of regional power supply inadequacy under such events.

Many studies have focused on exploring the distribution characteristics of RD events. The frequency and duration of RD events for onshore wind power in Germany have been analyzed using 40 years of reanalysis data and open-source software [10]. Case studies indicate that the longest recorded event lasted nearly ten days. An RD event in South Australia was analyzed using AEMO data from June 9 to June 12, 2021 [11]. Hourly reanalysis climatological data were utilized to analyze global historical trends in defined extreme shortage events from 1980 to 2022 [12]. That inefficient resource utilization was influenced by RD weather is indicated for China in [13], with risks concentrated in autumn and winter in regions such as Xinjiang, Sichuan, and the Yangtze River Basin. The combination of wind and solar power generation has been proposed to solve continuous RD problems in India [14].

As for risk assessment of such inadequacy, current studies mainly focus on extreme weather events and can be mainly categorized into two perspectives. First, from a demand-side perspective, specific weather conditions such as heatwaves [15] or snowstorms [16] can directly result in sharp increases or decreases in the temperature, leading to rapid growth in load demand and inadequacy in the power grid. A data-driven inverse modeling approach is proposed in [17] to inform residents to prepare for pre-cooling during heatwaves. Second, from a resilience perspective, typical extreme events such as hurricanes [18], blizzards [19], or wildfires [20] usually cause damage to the power system infrastructure, resulting in long-duration power shortages. A stochastic programming approach is proposed to increase the resiliency of a distribution system exposed to an approaching wildfire [21]. Resilience assessment and improved planning strategies have been discussed in [22] in cyber-physical power systems under typhoon disasters.

In contrast, RD events will mostly neither impact demand nor directly damage the power grid infrastructures. The spatiotemporal distribution of RD events also differs from the above-mentioned extreme weather events, such as hurricanes. Thus, quantification of long-term inadequacy risk and options for its reduction under RD events must differ from the studies outlined above. Moreover, most inadequacy problems under RD events result from a lack of regional flexibility resources [23]. Such long-term inadequacy implications warrant an integrated approach considering strategic and operational planning; Otherwise, it can lead to myopic solutions that could enhance power supply shortages.

On this basis, Table I further compares current studies on RD events in power systems. As shown in the table, most current studies primarily focus on the spatiotemporal analysis of RD events. Although many studies address energy balance within the RD scenario, few quantify inadequacy risk in planning. A critical gap, therefore, is the lack of a planning framework that represents RD-induced inadequacy risk and integrates it with traditional typical-day-based planning to support risk-aware expansion and flexibility enhancement decisions. Such a framework must account for the probability distribution of RD events and their associated power supply inadequacy, while consistently linking these extreme conditions with typical operating states. Accordingly, this paper develops such a framework and applies it to evaluate how different flexibility resources mitigate long-term inadequacy risk under RD events. While the underlying optimization model builds upon established formulations, the key technical contribution of this work lies in the structural representation of renewable drought risk and its integration into long-term planning, rather than in developing new optimization algorithms.

The main contributions of this paper are twofold:

1) Developing a meteorology-driven framework for renewable drought scenario construction that explicitly represents drought-induced risk arising from multi-day persistence and cross-regional dependence of renewable deficits;
2) Formulating a risk-aware co-planning model that integrates typical-day operation with renewable-drought events of varying durations to preserve inter-day flexibility requirements in investment decisions.

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TABLE I
COMPARISON OF RECENT STUDIES ON RENEWABLE DROUGHT RISKS IN POWER SYSTEMS

Study	Field	Main findings	Flexibility resources			Inadequacy Risk Quantification	Location
			Generation	Transmission	Storage		
[3]	SA	Examine climate change impacts on Europe's solar, wind, and hybrid energy droughts.	✓	×	×	no	Regional, Europe
[10]	SA	Analyzes the frequency, duration, and seasonal patterns of low-wind-power events in Germany over 40 years using reanalysis data.	×	×	×	no	Region Germany
[14]	SA	Examines how wind-solar hybrids reduce RD risks in India.	✓	✓	×	no	Regional, India
[24]	SA	Evaluate energy drought risks in a hydro-wind-PV system, emphasizing resource complementarity without modeling load balance.	✓	×	×	no	Local, Southwest China
[25]	SA	Assess climate-driven risks to wind power supply stability,	×	×	×	no	Global
[26]	SA	Reveals a trilemma of renewable energy quality—availability, variability, and extremeness—through wind and solar drought analysis in China.	✓	×	×	no	Regional, China
[8]	AA	Assess future RD risks in China under climate change using CMIP6 scenarios, focusing on supply-demand imbalances and regional vulnerabilities.	✓	×	×	no	Regional, China
[27]	AA	Studies how renewable resource droughts affect grid reliability and proposes measures to improve resilience.	×	×	×	no	ERCOT
[28]	EP	Evaluate climate impacts on wind–solar power systems and strategies to enhance their resilience.	✓	✓	✓	no	Global
[29]	EP	Propose a three-stage optimization model for co-planning of storage sizing, siting, and transmission expansion to ensure supply adequacy under severe renewable droughts.	✓	✓	✓	no	IEEE RTS-79 system
This paper	SA, AA & EP	Quantify and incorporate the inadequacy risk with RD scenarios into planning model; evaluate the contribution of each type of flexibility resources.	✓	✓	✓	yes	East China Power Grid

Note: In this paper, the current research fields are categorized into three aspects: Spatiotemporal Analysis (SA), Adequacy Assessment (AA), and Expansion Planning (EP). SA focuses on spatiotemporal analysis of RD events based on the real-world climate data. AA focuses on the energy balance and inadequacy risk of the power system under RD events. EP focuses on exploring how to implement flexibility resources to mitigate the RD risks.

The remainder of the paper is structured as follows. Section II provides the methodology to define and analyze the distribution of RD events; Section III establishes the CVaR-based risk quantification model; Section IV incorporates the CVaR-based risk into the power system planning model; Section V performs two case studies on the Garver's 6-node system and China's East Power Grid. Section VI concludes.

II. SPATIOTEMPORAL ANALYSIS OF RENEWABLE DROUGHT EVENTS

This section first introduces the basic analytical assumptions and then explains how RD events are defined in this paper. Finally, a spatiotemporal analysis of RD events is performed.

A. Basic Analysis Assumptions

1) The entire system is divided into several regions based on meteorological data, with each region containing at least one bus node. All bus nodes within the region share the same wind and solar curves
2) There are generally two ways to define RD events [10]: Constantly Below Threshold (CBT), which defines RD events that last two or more consecutive days [14]; and Mean Below Threshold (MBT) events, indicating that the mean renewable output over two or more consecutive days is lower than the threshold value [10]. This paper chooses CBT as the benchmark to define and select RD events.
3) In multiple regions, if RD occurs in any one region, the corresponding time series during that period will be selected for all regions.
4) This paper focuses on the inter-day long-term balance because existing intra-day flexibility resources, such as batteries, have already been widely deployed in power sectors, making the intra-day short-term power balance relatively well addressed. Therefore, this paper considers the RD scenario at daily resolution to explore the inter-day flexibility problems.

B. Definition of Renewable Drought Events

Equations (1)-(2) give a detailed definition of the renewable drought scenario. Equation (1) calculates the daily average renewable output $\omega_{r,d}$ based on the hourly renewable output $\omega_{r,t}$. $\mathcal{T}_d$ is the set of 24 hours within day d, while $ave(\cdot)$ is the statistical mean function. Equation (2) defines the renewable drought scenario s within region r. There are two key parameters to determine if the target scenario is an RD scenario: κ is the threshold ratio; δ represents the consecutive duration of the RD event, meaning that the target scenario should last at least δ days to be considered an RD event. $[\underline{d}_s, \overline{d}_s]$ represents the duration of renewable drought events. $\widehat{\omega}$ is the maximum value of renewable output during all sampling years.

$$\omega_{r,d} = \underset{t \in \mathcal{T}_d}{\text{ave}} \left(\omega_{r,t}\right) \tag{1}$$

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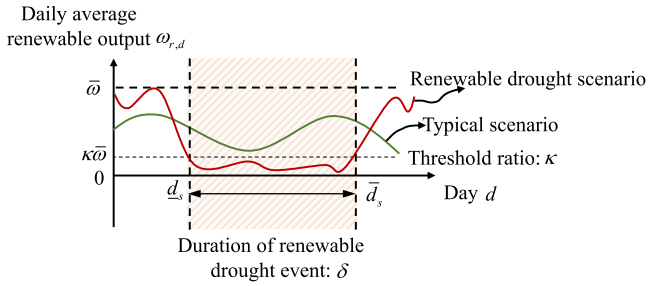


Fig. 1. Renewable drought scenario definition.

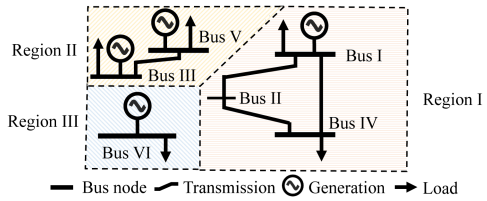


Fig. 2. Garver's 6-node system.

$$\exists r, s: \begin{cases} \omega_{r,d}\Big|_{d\in[\underline{d}_s,\bar{d}_s]} \le \kappa\widehat{\omega} \\ \bar{d}_s - \underline{d}_s \ge \delta \end{cases} \tag{2}$$

The renewable drought and typical renewable output scenarios are compared in Fig. 1, in which the shaded area illustrates the renewable drought scenario s starting from $\underline{d}_s$ and ending at $\bar{d}_s$. During the RD scenario, renewable output remains much lower than in the typical scenario, presenting significant challenges to the supply adequacy of a renewable-dominant power system.

C. Spatiotemporal Analysis

This subsection performs spatiotemporal analysis of RD events based on 44-year historical (1980-2023) ERA5 reanalysis hourly data [30]. This dataset, which underpins the analysis, provides hourly estimates of key atmospheric climate variables, including wind speed and solar radiation. Meteorological data is converted to renewable electricity output using wind turbine and solar panel power curves. This output is then applied to the modified Garver's 6-node system, which uses China's meteorological data for spatiotemporal analysis, as illustrated in Fig. 2. For comprehensive analysis, the modified system is divided into three interconnected regions according to the distribution of meteorological data. The three regions include two renewable energy bases in northwest China—the Kubuqi Desert and the Tengger Desert—and the Beijing power grid in the east. These are termed Regions I, II, and III, respectively. These regions were selected because they are already planned for interconnection via HVDC transmission. This makes spatiotemporal analysis of RD events across them essential. Table II shows the location of three regions.

In this paper, the threshold ratio κ is set to 0.1, following [10]. The choice does not affect the generality of our framework, as other values could also be used without altering the analysis method. To emphasize the inter-day adequacy problem—rather than the more extensively studied intra-day flexibility—we focus on RD events with durations δ of at least two days in the subsequent discussion.

TABLE II
LOCATION OF THREE REGIONS

Region	I	II	III
Longitude	109.3° E	107.6° E	116.4° E
Latitude	40.3° N	40.4° N	39.9° N

TABLE III
LONGEST DURATION OF RENEWABLE DROUGHT EVENT WITHIN EACH REGIONS AND SEASONS IN TERM OF DAYS

	Spring (Mar. 1-May 31)	Summer (June. 1-Aug. 31)	Autumn (Sep. 1-Nov. 30)	Winter (Dec. 1-Feb. 28)
Region I	5	10	10	11
Region II	6	6	13	10
Region III	6	16	7	7

The spatiotemporal distributions of RD events vary in different regions and seasons. Equation (3) calculates the average renewable output during the RD event $[\underline{d}_s, \bar{d}_s]$. Equation (4) formulates the matrix vector of the renewable output within each region. $\omega^{\mathrm{rd}}_{r,s}$ represent the average renewable output within region r.

$$\omega^{\mathrm{rd}}_{r,s} = \underset{d\in[\underline{d}_s,\bar{d}_s]}{\mathrm{ave}} \left(\omega_{r,d}\right) \tag{3}$$

$$\boldsymbol{\omega}_s = \left[\omega^{\mathrm{rd}}_{1,s}, \omega^{\mathrm{rd}}_{2,s}, \ldots, \omega^{\mathrm{rd}}_{N^{\mathrm{r}},s}\right]^{\mathrm{T}} \tag{4}$$

Table III depicts the longest duration of the RD events within different regions and seasons. The longest RD events within the three regions above occur in winter (Region I), autumn (Region II), and summer (Region III), respectively. Spring has relatively shorter RD durations (less than 6 days) in all regions. The maximum durations of the three regions are 11, 13, and 16, respectively.

To further investigate the spatiotemporal distribution characteristics of RD events, this paper proposes three indicators for quantification as follows:

1) RD Coincidence Factor τ_{ij}: Equation (5) defines the RD coincidence factor τ_{ij}, which measures the probability of RD events occurring simultaneously across regions i and j. This measure is based on the indicator function $\mathbb{I}(\cdot)$, which maps a condition to 1 if it is true and 0 if it is false, and the scenario duration δ_s, representing the length of the RD scenario s. $\mathbb{I}(\cdot)$ is the indicator function that assigns a value of 1 when the stated condition within the equation is true, and 0 otherwise. δ_s denotes the duration associated with the RD scenario s in the formula. The calculation of τ_{ij} depends on both the duration of each RD scenario δ_s and the total number of scenarios considered.

$$\tau_{ij} = \frac{\sum_s \delta_s \cdot \mathbb{I}\left(\left(\omega^{\mathrm{rd}}_{r=i,s} \le \kappa\widehat{\omega}\right) \wedge \left(\omega^{\mathrm{rd}}_{r=j,s} \le \kappa\widehat{\omega}\right)\right)}{\sum_s \delta_s \cdot \mathbb{I}\left(\left(\omega^{\mathrm{rd}}_{r=i,s} \le \kappa\widehat{\omega}\right) \vee \left(\omega^{\mathrm{rd}}_{r=j,s} \le \kappa\widehat{\omega}\right)\right)} \tag{5}$$

2) Weighted Pearson Correlation Coefficient ρ_{ij}: The weighted Pearson correlation coefficient ρ_{ij} is introduced to

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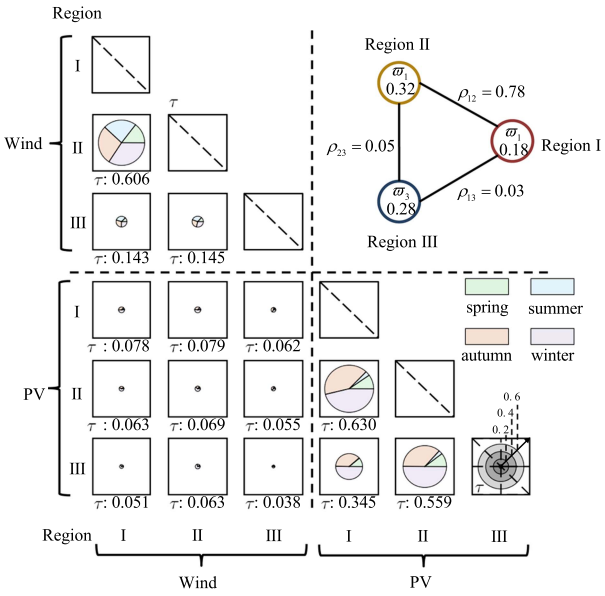


Fig. 3. Spatiotemporal distribution of renewable drought events.

measure the strength of the linear relationship between two regions i and j while incorporating the duration length δ_s of the RD event s as weights to account for the varying significance of each data point. Equation (6) computes the weighted average renewable output $\bar{\omega}_r^{\mathrm{rd}}$ throughout all RD scenarios. Equation (7) calculates the weighted Pearson correlation coefficient ρ_{ij}.

$$\bar{\omega}_r^{\mathrm{rd}} = \sum_s \delta_s \omega_{r,s}^{\mathrm{rd}} / \sum_s \delta_s \ \big|_{r\in\{i,j\}} \tag{6}$$

$$\rho_{ij} = \frac{\sum_s \delta_s \left(\omega_{i,s}^{\mathrm{rd}} - \bar{\omega}_i^{\mathrm{rd}}\right)\left(\omega_{j,s}^{\mathrm{rd}} - \bar{\omega}_j^{\mathrm{rd}}\right)}{\sqrt{\sum_s \delta_s \left(\omega_{i,s}^{\mathrm{rd}} - \bar{\omega}_i^{\mathrm{rd}}\right)^2 \sum_s \delta_s \left(\omega_{j,s}^{\mathrm{rd}} - \bar{\omega}_j^{\mathrm{rd}}\right)^2}} \tag{7}$$

3) Seasonal Gini Coefficient ϖ: The seasonal Gini coefficient ϖ quantifies the seasonal imbalance of RD events. Equations (8)-(9) calculate the frequency of RD events across different seasons. $[\chi_{\mathrm{sp}}, \chi_{\mathrm{su}}, \chi_{\mathrm{au}}, \chi_{\mathrm{wi}}]$ indicates the frequencies during spring, summer, autumn and winter respectively. Equation (10) computes the seasonal Gini coefficient, where x and y are the index of different seasons.

$$\boldsymbol{\chi} = [\chi_{\mathrm{sp}}, \chi_{\mathrm{su}}, \chi_{\mathrm{au}}, \chi_{\mathrm{wi}}]^{\mathrm{T}} \tag{8}$$

$$\chi_x = \sum_{s\in x} \delta_s / \sum_x \sum_{s\in x} \delta_s \ , x \in \{\mathrm{sp,su,au,wi}\} \tag{9}$$

$$\varpi = \frac{1}{2} \sum_{x,y} |\chi_x - \chi_y|, \forall x, y \in \{\mathrm{sp,su,au,wi}\} \tag{10}$$

Based on the above-proposed indicators, Fig. 3 further represents the spatiotemporal distributions of wind and solar drought events across Regions I-III. As shown in the upper right panel, each number in the circle represents the seasonal Gini coefficient ϖ, while each number on the line between the two circles is the Pearson correlation coefficient ρ_{ij}. Region II has the highest Gini coefficient of 0.32, indicating a relatively severe seasonal imbalance problem compared to the other regions. The time correlation of renewable output is stronger in Regions I and II with ρ_{12} of 0.78.

For the simultaneity analysis of RD events, the pie chart in each cell represents the simultaneous occurrence rate of RD conditions for wind and solar in different regions. The four colors indicate the proportion of simultaneous RD events across the four seasons, defined in the legend at right. The radius of each pie chart reflects the renewable output complementarity between regions during the RD events. For instance, the second row of the first column shows that the simultaneous occurrence rate of wind drought events within Regions I and II is 0.606, with the highest proportion occurring in winter. Region I and II have a higher coincidence factor for wind drought analysis, indicating a comparatively high spatial correlation. Simultaneous solar drought is more pronounced across the three regions, with coincidence factor τ as 0.345,0.559, and 0.630, respectively. The lower left section shows the correlation between wind and solar, while most values of τ are below 0.08, indicating a strong wind-solar complementarity within and across the regions. Appendix A further presents the confidence intervals for the RD coincidence factor (τ_{ij}), the weighted Pearson correlation coefficient (ρ_{ij}), and the seasonal Gini coefficient (ϖ_{ij}) at 95% confidence level for uncertainty analysis.

Based on the above analysis, it can be concluded that relying on a single flexibility resource to address the long-term inadequacy risk under the RD scenario is insufficient. Instead, inter-regional transmission, long-duration storage (LDES), and combining wind and solar generation could provide a portfolio of flexibility solutions to supply inadequacy episodes in renewable-dominant power systems and manage the risk from spatial, temporal, and cooperative perspectives.

III. Calculation of CVaR-Based Long-Term Inadequacy Risk

This section quantifies the risk of long-term inadequacy within each region under RD events. A multi-regional CVaR model is introduced for risk modeling. Algorithm 1 describes the CVaR-based regional risk calculation process as further outlined below.

A. Operation Simulation

Equations (11)-(18) perform the operation simulation under the RD scenario s. Equation (11) formulates the constraints on the hourly power balance. The generation set $\mathcal{G}_n$:$\{\mathcal{T}_n, \mathcal{W}_n, \mathcal{V}_n\}$ refers to generation type including thermal, wind and PV generation. $\mathcal{B}_n$:$\{\mathcal{X}_n, \mathcal{Y}_n\}$ is the storage connected to the bus node n, while $\mathcal{X}_n$ and $\mathcal{Y}_n$ are sets of short-duration battery storage and long-duration storage respectively. Equation (12) sets the upper and lower limit of the load curtailment. Equation (13) sums the load curtailment in scenario s as $\delta_{n,s,t}^{\mathrm{rd,cur}}$. Constraints (14) formulate the linearized unit commitment (UC) problem, including the output limit, ramp limit, and the startup/shutdown limit of thermal units. Detailed explanations of the linearization of the UC model can be seen in [31].

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Algorithm 1: CVaR-Based Risk Calculation.

Data input: $N \times 8760$ historical hourly renewable time series, RD threshold ratio κ, duration δ, power system topology data
for $y = 1$ **to** N **do**
  Calculate the daily average renewabel time series $\omega_{r,d}$
  for $d = 1$ **to** 365 **do**
    if $\omega_{r,d}\big|_{d\in[\underline{d}_s,\overline{d}_s]} \le \kappa\widehat{\omega}$ **and** $\overline{d}_s - \underline{d}_s \ge \delta$ **then**
      Select the target scenario as RD scenario and incorporate it into the RD scenario database: $s = s + 1$
      Perform hourly simulation and calculate the total load curtailment $\delta^{\text{rd,cur}}_{r,s}$ within the selected RD scenario s
    end if
  end for
end for
Analysis and statistics of the distribution of RD events based on metrics including RD coincidence factor τ_{ij}, Pearson correlation coefficient ρ_{ij} and seasonal Gini coefficient ϖ.
Decompose the inadequacy risk of the entire system into risks within each region r.
Formulate the distribution of load curtailment $\delta^{\text{rd,cur}}_{r,s}$ within each region r and RD scenario s.
Introduce CVaR method to quantify the regional inadequacy risk $\text{CVaR}_r(\delta^{\text{rd,cur}}_{r,s}, \alpha)$ under the confidence level α within each region r.

Using a transportation model, (15) formulates the power flow constraints for transmission line l. The transportation model for power flow is widely used in current regional power system planning [32]. Equation (16) shows the output limits of renewable generation. $\mathcal{W}_r$ and $\mathcal{V}_r$ are sets of wind and PV units placed in region r. $p^{\text{rd}}_{g,s,t}$ and $p^{\text{rd,cur}}_{g,s,t}$ are the renewable output and curtailment respectively. Constraints (17) formulate the operating constraints for short- and long-duration storage in the RD scenario s, including the inventory change between the adjoining hour, the upper/lower limits of the charging/discharging power, and the inventory. Equation (18) depicts the energy balance of the storage inventory for short-duration battery storage set $\mathcal{X}$ within daily operation cycle, where t_{start} and t_{end} are starting/ending hours of each day in RD scenario s. It is assumed that LDES operates within a yearly cycle; thus, the start and end inventory of LDES during the RD scenario are linked via charging/discharging behaviors and do not need to be equal.

$$\sum_{g\in\mathcal{G}_n} p^{\text{rd}}_{g,s,t} + \sum_{b\in\mathcal{B}_n}\left(p^{\text{dis,rd}}_{b,s,t} - p^{\text{cha,rd}}_{b,s,t}\right) + \sum_{l\in\mathcal{L}^+_n} f^{\text{rd}}_{l,s,t} - \sum_{l\in\mathcal{L}^-_n} f^{\text{rd}}_{l,s,t} = d^{\text{rd}}_{n,s,t} - d^{\text{cur,rd}}_{n,s,t}, \forall n,s,t \tag{11}$$

$$0 \le d^{\text{cur,rd}}_{n,s,t} \le d^{\text{rd}}_{n,s,t}, \forall n,s,t \tag{12}$$

$$\delta^{\text{rd,cur}}_{r,s} = \sum_{n\in\mathcal{N}_r,t} d^{\text{rd,cur}}_{n,s,t} \tag{13}$$

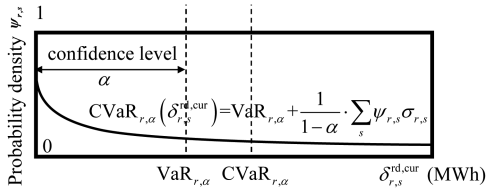


Fig. 4. CVaR-based calculation method.

$$\begin{cases} 0 \le \lambda_g o^{\text{rd}}_{g,s,t} \le p^{\text{rd}}_{g,s,t} \le o^{\text{rd}}_{g,s,t} \le G_g + G^0_g \\ -\gamma^{\text{dn}}_g o^{\text{rd}}_{g,s,t} \le p^{\text{rd}}_{g,s,t} - p^{\text{rd}}_{g,s,t-1} \le \gamma^{\text{up}}_g o^{\text{rd}}_{g,s,t} \\ o^{\text{rd}}_{g,s,t} = o^{\text{rd}}_{g,s,t-1} + o^{\text{rd,up}}_{g,s,t} - o^{\text{rd,dn}}_{g,s,t} \\ o^{\text{rd}}_{g,s,t} \ge \sum_{\tau=1}^{T^{\text{on}}_g} o^{\text{rd,up}}_{g,s,t-\tau} \\ o^{\text{rd}}_{g,s,t} \le G_g + G^0_g - \sum_{\iota=1}^{T^{\text{off}}_g} o^{\text{rd,dn}}_{g,s,t-\iota} \\ \forall g \in \mathcal{T}, s, t \end{cases} \tag{14}$$

$$\left|f^{\text{rd}}_{l,s,t}\right| \le L_l + L^0_l \tag{15}$$

$$\begin{cases} 0 \le p^{\text{rd}}_{g,s,t} \le \left(G_g + G^0_g\right)\omega^{\text{rd}}_{r,s,t} \\ p^{\text{rd}}_{g,s,t} + p^{\text{rd,cur}}_{g,s,t} = \left(G_g + G^0_g\right)\omega^{\text{rd}}_{r,s,t} \\ \forall \text{g} \in \{\mathcal{W}_r, \mathcal{V}_r\}, r, s, t \end{cases} \tag{16}$$

$$\begin{cases} q^{\text{rd}}_{b,s,t} = q^{\text{rd}}_{b,s,t-1} + \eta^{\text{cha}}_b p^{\text{cha,rd}}_{b,s,t} - p^{\text{dis,rd}}_{b,s,t}/\eta^{\text{dis}}_b \\ 0 \le p^{\text{cha,rd}}_{b,s,t}, p^{\text{dis,rd}}_{b,s,t} \le B_b + B^0_b \\ 0 \le q^{\text{rd}}_{b,s,t} \le T_b\left(B_b + B^0_b\right) \\ b \in \{\mathcal{X}, \mathcal{Y}\}, s, t \end{cases} \tag{17}$$

$$q^{\text{rd}}_{b,s,t_{\text{start}}} = q^{\text{rd}}_{b,s,t_{\text{end}}}, b \in \mathcal{X} \tag{18}$$

It should be noted that the linearized transportation flow and clustered UC model in this paper just reasonably simplify some physical constraints to improve the computation efficiency. While such kinds of simplification might still overestimate the thermal unit flexibility or power transmission to some extent. Future work might still need to focus on the detailed modeling of these constraints to accurately describe the multi-day power supply scarcities under RD events.

B. CVaR-Based Risk Quantification

Fig. 4 shows the calculation of the Conditional Value at Risk (CVaR)-based risk in detail. The Value at Risk (VaR) calculates the expected power shortage in a renewable scenario under confidence level α. CVaR considers the mean value greater than the VaR cutoff point. As shown in the figure, $\psi_{r,s}$ represents the weight of RD scenario s within region r, describing the distribution of the RD events. Thus, the probability of RD events in our planning model is determined via the statistics of regional RD events based on the historical data. The specific calculation is formulated in Appendix B, (B.1)-(B.8). Detailed mathematical derivation could be found in [33].

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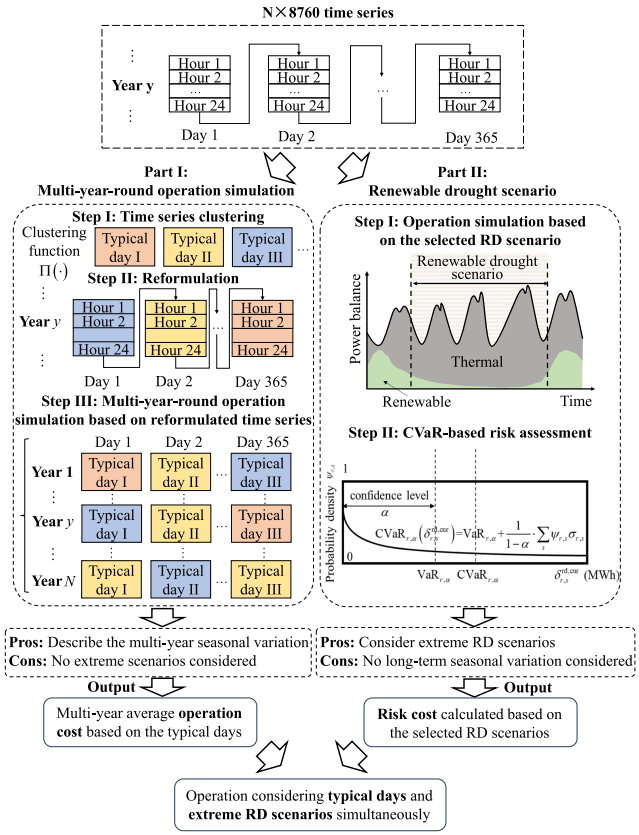


Fig. 5. Framework of operation simulation based on multi-year-round time series.

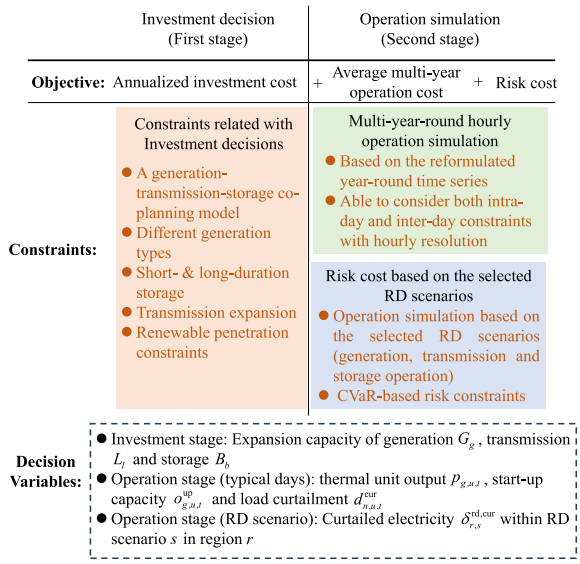


Fig. 6. Framework of the planning model.

IV. Power System Planning Model Considering Long-Term Inadequacy Risk

The traditional planning method usually simulates the power system operation within selected typical days or weeks instead of a year-round hourly time series for purposes of computational efficiency. However, the selected typical scenarios cannot adequately capture long-term variations across seasons or describe scarcities under extreme scenarios such as RD events. The main challenges lie in the inability to simultaneously describe the long-term seasonal fluctuation characteristics and the operating states under extreme scenarios based on an $N \times 8760$-hour multi-year time series, while ensuring the computational efficiency of the optimization.

A. Simulation Framework Based on Multi-Year Time Series

Fig. 5 illustrates the simulation details, including two parts based on the $N \times 8760$ multi-year time series. In Part I, the $N \times 8760$ time series are first clustered into several typical days based on the clustering function $\Pi(\cdot)$ formulated in (19). $\Pi(\cdot)$ indicates the mapping relation from natural day d to the typical day u. This paper adopts the K-medoids method for clustering. Based on the mapping relationship, the clustered typical days are then reformulated as multi-year-round time series. Consequently, long-term coupled constraints such as LDES operation can be modeled chronologically. The detailed formulation process can be further seen in [34]. In Part II, the CVaR-based inadequacy risk is evaluated as indicated in Section III. The following section further explains the formulation details of Part I and the objective function in the planning model.

$$u = \Pi(d) \tag{19}$$

B. Formulation Details

The planning model is depicted in Fig. 6 as an optimization problem. This paper considers the expansion planning of generation, transmission, and storage in the planning stage. In the operation simulation stage, the operation constraints include multi-year operation constraints and risk constraints. The multi-year operation constraints can be further categorized into intra-day and inter-day constraints:

1) Objective Function: Equations (20)–(24) formulate the objective function. Equation (20) is the total cost including the annualized investment cost C^{inv}, the average yearly operation cost C^{ope} and inadequacy risk cost C^{risk}. Equation (21) calculates the annualized investment cost, including the investment in generation units G_g, transmission line L_l, and storage B_b. ρ is the discount rate and p is the investment payback year. Equation (22) is the annualized operation cost, including the fuel and start-up cost of coal-fired generation and the load curtailment cost. As the output of the clustering function $\Pi(\cdot)$ in (19), π_u is the weight of typical day u throughout N years, assigning different weights to each typical day. Equation (23) shows that the sum of all weights π_u equals the total number of calendar days in the historical period, ensuring comprehensive coverage over time. Equation (24) sums the long-term inadequacy risk cost within RD scenarios of each region r.

$$\min C^{\text{ove}} = C^{\text{inv}} + C^{\text{ope}} + C^{\text{risk}} \tag{20}$$

$$C^{\text{inv}} = \left((1+\rho)^p - 1\right) / \rho(1+\rho)^p \cdot$$

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$$\left(\sum_{g} c_g^{\text{inv}} G_g + \sum_{c} c_l^{\text{inv}} L_l + \sum_{b} c_b^{\text{inv}} B_b\right) \tag{21}$$

$$C^{\text{ope}} = \sum_{u,t} \pi_u \left(\begin{array}{l} \sum_{g} \left(c_g^{\text{ope}} p_{g,u,t} + c_g^{\text{up}} o_{g,u,t}^{\text{up}}\right) \\ + \sum_{n} c_n^{\text{cur}} d_{n,u,t}^{\text{cur}} \end{array} \right) / N \tag{22}$$

$$\sum_{u} \pi_u = N \times 365 \tag{23}$$

$$C^{\text{risk}} = \sum_{r} c_r^{\text{cur}} \text{CVaR}_{r,\alpha}\left(\delta_{r,s}^{\text{rd,cur}}\right) \tag{24}$$

2) Intra-Day Operation Constraints: Intra-day operational constraints define the power system's performance for each typical day u. Constraints within each typical day are treated independently. Most intraday constraints are formulated similarly to those in (11)-(18) for the RD scenario. As shown in (25), these constraints include the node power balance, load curtailment, linearized unit commitment [31], power flow, renewable output and battery storage operation constraints. The primary distinction is that these constraints and variables are defined within the typical day u rather than the RD scenario s, thereby maintaining consistency across the formulation. To streamline the discussion, detailed modeling is omitted here. Instead, detailed formulations are presented in Appendix C.

$$\forall u: \text{constraints} \begin{cases} \text{power balance,load shedding} \\ \text{linearized UC,power flow} \\ \text{renewable output, battery storage} \end{cases} \tag{25}$$

3) Inter-Day Operation Constraints: The inter-day constraints describe the cross-daily operation constraints of power systems. This paper mainly focuses on the cross-daily operation constraints of LDES.

$$\begin{cases} q_{b,d,t} = q_{b,d}^{\text{inter}} + q_{b,d,t}^{\text{intra}} \big|_{u=\Pi(d)} \\ q_{b,d,t=1}^{\text{intra}} = 0 \\ \Delta q_{b,d}^{\text{intra}} = \sum_{t=1}^{t_{\text{end}}^{\text{d}}} \left(\eta_b^{\text{cha}} p_{b,u,t}^{\text{cha}} - p_{b,u,t}^{\text{dis}} / \eta_b^{\text{dis}}\right) \big|_{u=\Pi(d)} \\ q_{b,d}^{\text{inter}} = q_{b,d-1}^{\text{inter}} + \Delta q_{b,d}^{\text{intra}} \\ 0 \le p_{b,u,t}^{\text{cha}}, p_{b,u,t}^{\text{dis}} \le B_b \\ 0 \le q_{b,d,t} \le T_b B_b \\ q_{b,1,1} = q_{b,d_{\text{end}},t_{\text{end}}^{\text{d}}} \\ \forall b \in \mathcal{Y}, d, t \end{cases} \tag{26}$$

Equation (26) defines the operation constraint set of LDES. It covers the inventory change between adjacent hours, charging/discharging power limits, and storage inventory limits. As LDES operates in yearly cycles, modeling $q_{b,d,t}$ should capture hourly inventory changes over the year. Therefore, the hourly inventory $q_{b,d,t}$ of LDES b is decomposed into two parts: the inter-day part $q_{b,d}^{\text{inter}}$ and intra-day part $q_{b,u,t}^{\text{intra}}$. $q_{b,d}^{\text{inter}}$ is the storage inventory at the start of natural day d, describing daily variation. $q_{b,u,t}^{\text{intra}}$ shows the hourly variation within the typical day u that is mapped with natural day d. Note that $q_{b,u,t=1}^{\text{intra}}$ in the second row is 0. The third row of (26) calculates the net change of storage inventory $\Delta q_{b,d}^{\text{intra}}$ within the typical day u, after considering charging/discharging in day u. The fourth row links inventory between consecutive natural days. The fifth and sixth rows constrain charging/discharging power and storage inventory, with T_b as storage duration. The last equation sets the yearly balance constraint. At the start and end of the calendar year, d_{end} and $t_{\text{end}}^{\text{d}}$ mark the final day of the year and final hour of the day. Detailed modeling discussions can be found in [34].

4) Risk Constraints Under RD Events: The inadequacy risk constraints formulated in (B.1)-(B.8) are incorporated into the planning model to quantify the long-term inadequacy risk of regional power supply under RD events.

Finally, (11)-(26) formulate the planning model. Among them, (20)-(24) establish the objective functions, while others are operation constraints. By linearizing the operation constraints, including the UC problem, the planning problem becomes a typical large-scale linear programming model that can be efficiently solved by commercial solvers (such as Gurobi). The convergence criterion is set to 1% to ensure the accuracy.

TABLE IV
SETTINGS OF DIFFERENT CASE STUDIES

	SC-I	SC-II	SC-III	SC-IV	SC-V
Typical day	✓	✓	✓	✓	✓
RD	×	✓	✓	✓	✓
Interconnection	×	×	✓	✓	✓
LDES	×	×	×	✓	✓
Hybrid	×	×	×	×	✓

V. CASE STUDIES

This section performs two case studies on 1) the modified Garver's 6-node system introduced in Section II-C and 2) China's East Power Grid in 2035.

Table IV compares different settings of five scenarios (SCs). As the table shows, SC-I is the widely used planning model. It only considers typical day scenarios selected from the 44-year historical (1980-2023) ERA5 reanalysis hourly data. SC-II first introduces CVaR-based risk measurement to prove the necessity of considering inadequacy risks under RD events in the planning problem. In comparison, SCs III-V are performed to explore the potential benefits of different flexibility measures in mitigating long-term inadequacy risk under the RD scenario. SC-III considers cross-regional transmission to strengthen regional interconnection in power supply. On this basis, SC-IV introduces hydrogen storage as long-duration energy storage (LDES) to enhance the power system's long-term flexibility. Finally, SC-V considers combining wind and PV generation, enabling hybrid renewable generation to exploit their potential temporal complementarities within the RD scenarios. All simulations were run using Gurobi 12.0 under Yalmip on a Windows-based server with 20 processors clocking at 3.7 GHz and 64 GB of RAM.

A. Garver's 6-Node System

The modified Garver's 6-node system is depicted in Fig. 2. The system is divided into three regions to investigate the spatiotemporal distribution of regional inadequacy risk. Twelve

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TABLE V
COMPARISONS OF RESULTS BASED ON GARVER'S 6-NODE SYSTEM

Scenario	I	II	III	IV	V
Thermal (MW)	441	950	721	353	415
Wind (MW)	2844	3440	2211	2003	811
PV (MW)	–	–	–	–	1216
Battery (MW)	1127	1102	563	348	528
Transmission(MW)	–	–	470	383	465
LDES (MW)	–	–	–	389	295
LCOE (CNY/kWh)	0.589	0.724	0.488	0.446	0.392
Modeling time (s)	5	9	13	17	24
Solving time (s)	32	688	1200	3147	4499
EENS (GWh/yr)	9.67	8.78	6.01	5.68	1.03

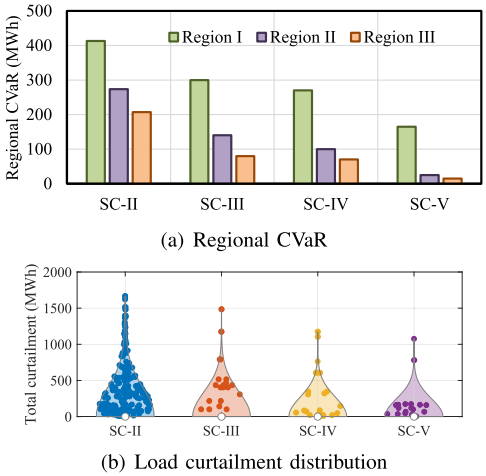
typical days are selected from a 44-year time series for optimal planning.

1) Comparisons of Planning Results: Table V compares the results of different case studies. All cases focus on the optimal generation, transmission, and storage configuration with the confidence level α set at 95%. Based on the optimization results, each case further performs a reliability analysis and calculates the expected energy not served (EENS) to test the reliability of each planning scheme. A 44-year-round operation simulation is performed for EENS calculation. While generator forced outages and maintenance were not included in the EENS calculation.

The comparison between SC-I and SC-II proves the necessity of introducing risk measurement in the planning problem: introducing risk constraints leads to a higher requirement for thermal generation, increasing from 441 MW (SC-I) to 950 MW (SC-II) to provide additional flexibility. The levelized cost of electricity (LCOE) has also increased from 0.589 CNY/kWh (SC-I) to 0.724 CNY/kWh (SC-II). As a result, the EENS drops from 9.67 GWh/yr to 8.78 GWh/yr in the reliability analysis.

In comparison, SCs III-V further explore the potential benefit of different flexibility measures in mitigating inadequacy risk. After introducing flexibility from the inter-regional transmission, the thermal installation drops from 950 MW (SC-II) to 721 MW (SC-III) with better reliability performance (2.77 GWh/yr reduction of EENS). In SC-IV, hydrogen storage is introduced as LDES to provide long-term flexibility from a temporal perspective. After installing 389 MW LDES, thermal generation and transmission expansion capacity decrease, with a 0.42 GWh/yr reduction of EENS. Finally, SC-V considers a diverse portfolio of wind and solar generation. By combining wind and PV generation, the planning scheme can realize a minimum EENS of 1.03 GWh/yr at an economical cost of 0.392 CNY/kWh. Based on the above comparative assessments, we have validated the necessity of introducing climate risk into the planning model to guarantee the reliability level. Moreover, we have quantified the contribution of different flexibility measures to improving long-term flexibility and mitigating the inadequacy risk, further highlighting the potential value of our proposed method.

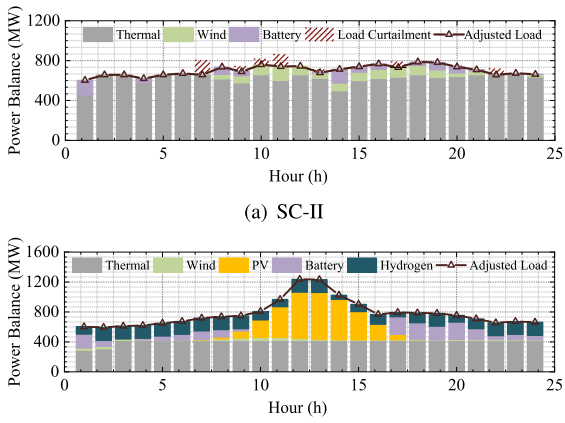
As for the computation time (modeling time + solving time), it should be noted that the computation of SC-I takes only 37 s; While in SC-V, the computation time increases to more than 4000 s. This is mainly because the introduction of massive RD scenarios significantly boosts the calculation complexity. Thus, when applied to a larger power grid, the calculation needs to do simplifications such as shrinking the number of typical days.



(a) Regional CVaR

(b) Load curtailment distribution

Fig. 7. Comparisons of risk distribution. (a) Regional CVaR (b) Load curtailment distribution.



(a) SC-II

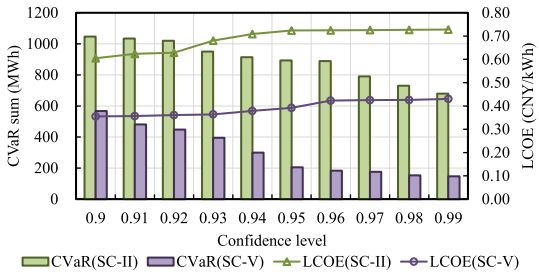
(b) SC-V

Fig. 8. Hourly power balance results. (a) SC-II (a) SC-V.

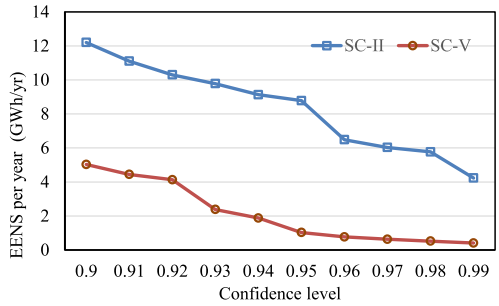
Fig. 7 compares the risk distributions in different cases. Fig. 7(a) shows the regional CVaR values. Region I has the highest inadequacy risk among the three regions. With the introduction of different flexibility measures, all regions' risks drop rapidly from 413, 273, 207 MWh (SC-II) to 165, 25, 15 MWh (SC-V). As depicted in Fig. 7(b), the violin plots depict the distribution of the curtailed electricity. Each dot in the figure represents the load curtailment within one RD scenario. Through the coordination of multiple types of flexibility resources, the number of load-shedding scenarios has been effectively reduced by over 90% from SC-II to SC-V. The maximum curtailed load electricity within each scenario across the system has decreased from 1723 MWh (SC-II) to 1107 MWh (SC-V).

Fig. 8 illustrates one representative day from a wind drought scenario to compare the balance results between SC-II and SC-V. In the figure, the adjusted load is defined as $d^{\mathrm{rd}}_{n,s,t} + p^{\mathrm{cha,rd}}_{b,s,t}$, where $d^{\mathrm{rd}}_{n,s,t}$ denotes the local native load and $p^{\mathrm{cha,rd}}_{b,s,t}$ represents the charging power of storage unit b. The discharging power is displayed in the figure as one type of generation, alongside

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(a) CVaR sum and LCOE



(b) EENS

Fig. 9. Sensitivity analysis with increasing confidence level. (a) CVaR sum and LCOE (b) EENS.

thermal, wind, and PV generation. Compared to SC-II, SC-V introduces all possible flexibility resources as listed in Table IV. In SC-II, load curtailment occurs from 7:00 to 12:00, 16:00 to 18:00, and 21:00 to 23:00, while most flexibility is provided via thermal generation and batteries. In comparison, introducing multiple flexibility measures in SC-V reduces the total load curtailment to zero in the same wind drought scenario. The surplus PV electricity at noon is stored in batteries and discharged during the sharp decrease of PV output in the afternoon to supply the load, significantly increasing the adjusted midday load demand. Although there is relatively low thermal generation installation compared to SC-II, the inter-regional electricity transmission, LDES, and the combined generation of both wind and PV provide sufficient flexibility.

2) Sensitivity Analysis: Fig. 9 represents sensitivity analysis with the increase of confidence level α ranging from 0.90 to 0.99. Fig. 9(a) depicts the change of the sum of regional CVaR and LCOE. The growth of α sets increasingly higher requirements for the risk management level, leading to a decreasing CVaR value with a higher LCOE value. As the confidence level reaches 99%, the CVaR value is 680 MWh in SC-II at the LCOE of 0.728 CNY/kWh. In comparison, SC-V lowers the LCOE to 0.430 CNY/kWh and has a much reduced CVaR value of 147 MWh. Fig. 9(b) illustrates the annual variation of EENS. An increase in the confidence level α imposes stricter requirements on risk mitigation, thereby significantly reducing the system's annual expected EENS.

B. China's East Power Grid

1) Description of the Test System: The proposed model in this paper is further applied to China's East power grid, as shown in Fig. 10. The system is divided into eight regions: five

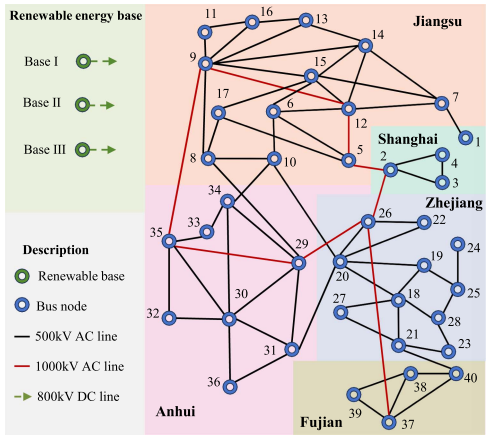


Fig. 10. China's East Power Grid.

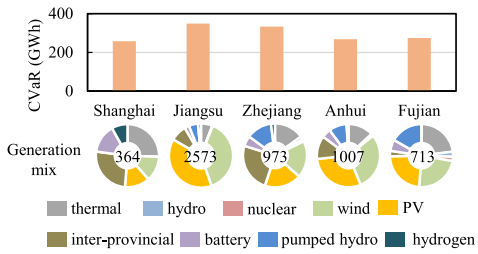
TABLE VI
LOCATION OF EAST CHINA POWER GRID

Region	Longtitude	Lagtitude
Shanghai	121.5° E	31.2° N
Jiangsu	118.8° E	32.1° N
Zhejiang	120.2° E	30.3° N
Anhui	117.3° E	31.9° N
Fujian	119.3° E	26.1° N
Base I	109.3° E	40.3° N
Base II	106.1° E	39.2° N
Base III	102.3° E	37.2° N

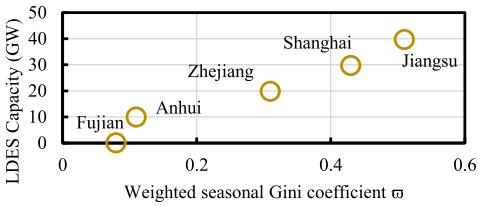
neighboring eastern provincial power grids and three remote renewable energy bases in northwest China. 800 kV HVDC candidate lines are to connect between remote northwestern renewable bases and eastern load centers; Most inter-provincial transmission lines are 500 kV AC lines, while the rest are 1000 kV HVAC transmission lines. This section selects 4 days out of a 44-year time series for optimal planning. While the computation time for the optimization takes more than 9000 s. Table VI depicts the location of five eastern provinces and three remote renewable energy bases. All meteorological data is collected and calculated based on the 44-year historical (1980-2023) ERA5 reanalysis hourly data within the corresponding location. This section mainly analyzes the inadequacy risk of the five eastern provincial grids as load centers. The detailed information for existing generation and storage within the grid is shown in Appendix D.

2) Discussions of Results: Fig. 11 analyzes the planning results based on assumptions in SC-V from Table IV. Fig. 11(a) compares regional CVaR under different provincial generation mixes. Jiangsu province has the highest CVaR value, at 348 GWh, with a total generation capacity of 2573 GW. Zhejiang and Shanghai have the highest proportion of imported electricity through inter-provincial transmission, both greater than 20%. Fig. 11(b), on the other hand, investigates the provincial installed LDES capacity with the corresponding provincial weighted seasonal Gini coefficient ϖ. As demonstrated in the figure, a larger Gini coefficient leads to a higher LDES installed

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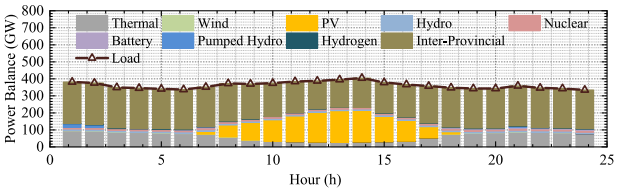


(a) Regional risk verses generation mix

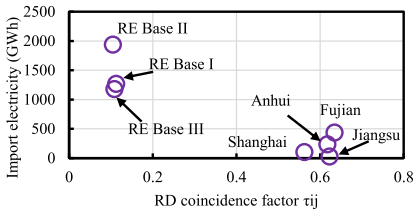


(b) Regional LDES capacity verses seasonal Gini coefficient

Fig. 11. Planning results analysis of China's East Power Grid.



(a) Hourly power balance of Zhejiang Province during RD events



(b) Imported energy from other regions verses coincidence factor during RD events

Fig. 12. Power balance of Zhejiang Province during RD events.

capacity, which indicates that increased LDES deployment significantly mitigates seasonal variations. Jiangsu province has the highest LDES capacity of 40 GW, with a Gini coefficient of 0.51.

As an example, the hourly operations during a day from a selected wind drought scenario for Zhejiang province are shown in Fig. 12(a). Among all the flexibility measures mentioned, inter-provincial transmission from other provinces is the dominant power supply resource, as it adds up to 5161 GWh during one day, showing its significant contribution to Zhejiang's energy supply under this scenario. On this basis, Fig. 12(b) depicts the imported electricity from all interconnected regional grids, including three renewable energy bases and four provincial grids under the corresponding RD coincidence factor τ_{ij} between the targeted province and Zhejiang province. The figure shows that

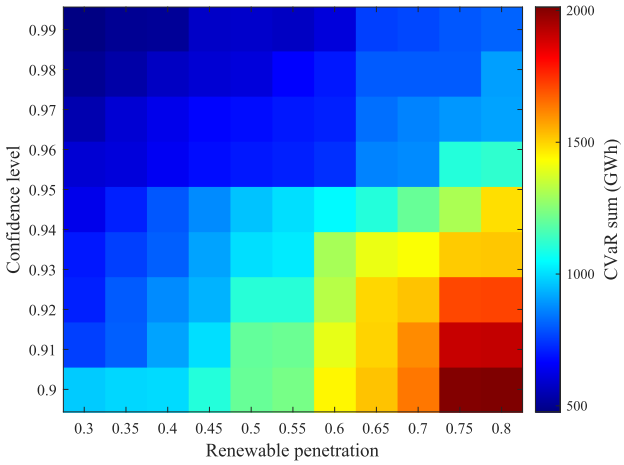


Fig. 13. Sensitivity analysis via renewable penetration and confidence level.

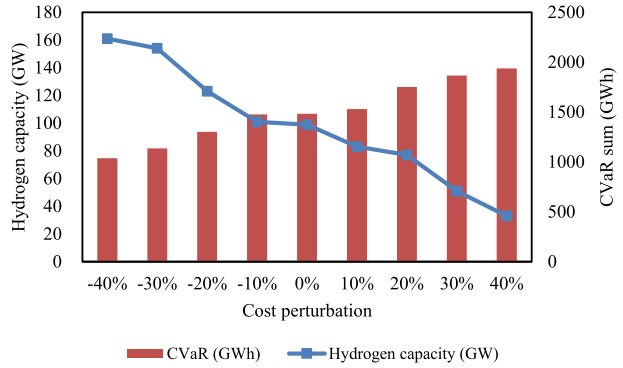


Fig. 14. Sensitivity analysis via the perturbation of hydrogen storage CAPEX.

the three RE bases could provide Zhejiang with more electricity than the other four interconnected load center provinces. Meanwhile, the RD coincidence factors of the three RE bases (0.113, 0.105, 0.1096) are lower than those of the other four provinces (0.563, 0.623, 0.618, 0.635). Thus, we conclude that a lower RD coincidence factor, indicating weaker simultaneity between two regions, provides more complementary inter-regional performance during the RD scenario by ensuring less correlated supply patterns.

Fig. 13 presents further sensitivity analysis based on the entire system with the increasing renewable penetration from 0.3 to 0.8 and confidence level from 0.90 to 0.99. The color range in the figure indicates the sum of the CVaR values in the entire system. The CVaR value reaches its highest value at 2177 GWh, with renewable penetration at 0.8 and a confidence level of 0.9. In conclusion, higher renewable integration requires a higher confidence level to keep the risk level consistent.

Fig. 14 performs sensitivity analysis on how the installed capacity of LDES in the system and the corresponding CVaR risk change as the CAPEX of LDES varies from 40% lower to 40% higher relative to the current level. As the figure depicts, the increase in the hydrogen CAPEX would reduce the hydrogen storage capacity, with the increase in the sum of CVaR risk.

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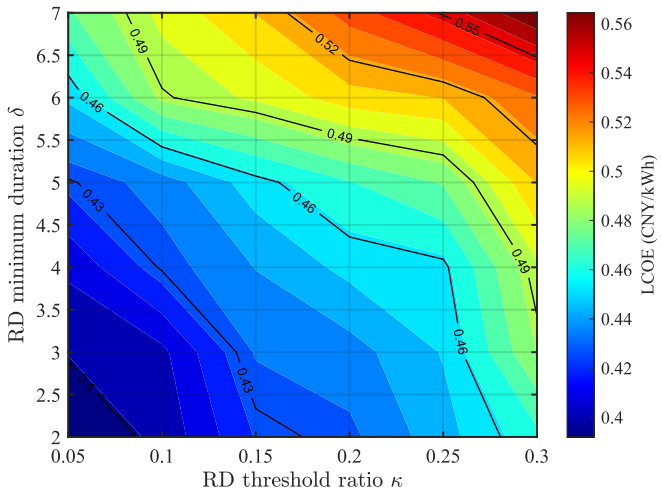


Fig. 15. Sensitivity analysis via the change of threshold ratio and minimum duration.

Fig. 15 depicts the change of LCOE with the increase of RD threshold κ (from 0.05 to 0.3 with 0.05 resolution) and minimum duration δ (from 2 to 7 days). As the figure shows, an increase in the κ and δ will both increase the LCOE of the whole system: A larger minimum duration δ intensifies CVaR's focus on tail extremes, while an increased κ raises scenario complexity, jointly leading to higher CVaR.

VI. Conclusion

This paper develops a framework for representing renewable-drought-induced inadequacy risk in long-term power system planning. By integrating meteorology-driven RD scenarios with typical-day operation in a risk-aware planning model, the proposed approach captures the planning implications of persistent and cross-regional renewable deficits. Case studies show that RD-induced risk varies significantly across regions and seasons, and that coordinated flexibility resources—especially inter-regional transmission, long-duration storage, and diversified wind–solar portfolios—can effectively mitigate such risk. Future work will extend the framework to compound climate extremes, examine the effect of different risk preferences on planning schemes, and further investigate enhanced risk-aware formulations, more efficient solution techniques, and relevant theoretical properties to improve computational tractability and scalability in large-scale planning applications.

Appendix A: Confidence Interval Calculation

Fig. A1 presents the confidence intervals for the RD coincidence factor (τ_{ij}), the weighted Pearson correlation coefficient (ρ_{ij}), and the seasonal Gini coefficient (ϖ_{ij}) at the 95% confidence level. The intervals were estimated using the block bootstrap method applied over yearly blocks. Specifically, the 95% confidence intervals were constructed from the 2.5th and 97.5th percentiles of the bootstrap distribution. The detailed procedure follows the approach described in [35].

TABLE VII
Existing Generation/Storage Capacity Within Each Region

Capacity (GW)	Shanghai	Jiangsu	Zhejiang	Anhui	Fujian
Coal	16.2	99.6	57.8	70.5	32.9
Gas	11.2	20.8	16	1.4	3.9
Hydro	0	0	1.8	0.2	3.1
Nuclear	0	6.7	10.3	0	14
Utility-scale PV	0	6.2	4	3.5	0
Distributed PV	0.5	31.6	27.2	33.1	3.5
Onshore wind	0	14.6	0.5	5.5	8.9
Offshore wind	1.6	16.2	5.9	0	1.6
Pumped hydro	0	3.5	9.8	5.6	5
Battery	0	0	0.7	0	0

Appendix B: CVaR-based Risk Calculation

Equations (B.1)-(B.8) present the calculation of long-term inadequacy risk using the CVaR model. The subsequent equations detail the decomposition, scenario probability derivation, normalization, and linearization steps involved in this process. Equation (B.1) decouples the inadequacy risk of the system into risks of each region r. This is based on the subadditivity of CVaR [36]. Here, $\mathrm{CVaR}_{r,\alpha}(\cdot)$ is the CVaR-based risk calculation function of region r, and α is the confidence level. Equation (B.2) defines the regional risk value $\mathrm{CVaR}_{r,\alpha}(\delta^{\mathrm{rd,cur}}_{r,s})$ where $\mathbb{E}(\cdot)$ denotes the expectation operator. Equation (B.3) outlines the calculation process using continuous integration, where $\psi_{r,s}$ is defined as the probability of the RD scenario for region r and scenario s. Equation (B.4) provides the derivation of $\psi_{r,s}$, the probability of scenario s for region r, based on the duration that each scenario occurs during the 44-year historical reconstruction. Equation (B.5) shows the sum of RD scenario probability $\psi_{r,s}$ equals 1. Equations (B.6)-(B.8) discretize the integration into a linear form. Equation (B.6) calculates the regional CVaR value. Equation (B.7) defines the value of $\sigma_{r,s}$, and (B.8) introduces the auxiliary variable $\sigma_{r,s}$ for linearization.

$$\mathrm{CVaR}_{\alpha}\left(\sum_{r}\delta^{\mathrm{rd,cur}}_{r,s}\right)\le\sum_{r}\mathrm{CVaR}_{r,\alpha}\left(\delta^{\mathrm{rd,cur}}_{r,s}\right) \tag{B.1}$$

$$\mathrm{CVaR}_{r,\alpha}\left(\delta^{\mathrm{rd,cur}}_{r,s}\right)\triangleq\mathbb{E}\left(\delta^{\mathrm{rd,cur}}_{r,s}\,\middle|\,\delta^{\mathrm{rd,cur}}_{r,s}\ge\mathrm{VaR}_{r,\alpha}\right) \tag{B.2}$$

$$\mathrm{CVaR}_{r,\alpha}\left(\delta^{\mathrm{rd,cur}}_{r,s}\right)=\mathrm{VaR}_{r,\alpha}+\frac{1}{1-\alpha}\int_{\delta^{\mathrm{rd,cur}}_{r,s}\ge\mathrm{VaR}_{r,\alpha}}\psi_{r,s}\left(\delta^{\mathrm{rd,cur}}_{r,s}-\mathrm{VaR}_{r,\alpha}\right)d\delta^{\mathrm{rd,cur}}_{r,s} \tag{B.3}$$

$$\psi_{r,s}=\delta_s/\sum_{s}\delta_s \tag{B.4}$$

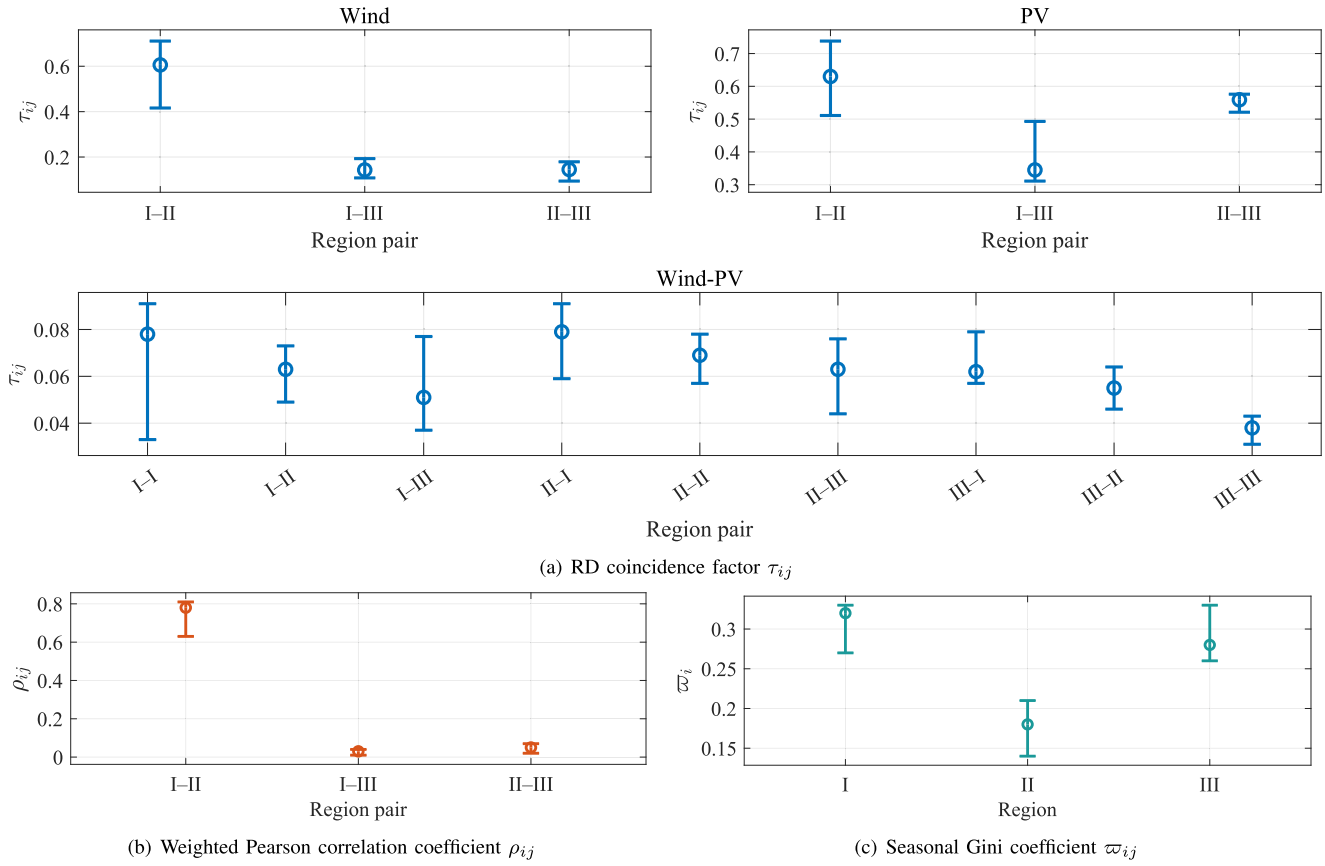
$$\sum_{s}\psi_{r,s}=1 \tag{B.5}$$

$$\mathrm{CVaR}_{r,\alpha}\left(\delta^{\mathrm{rd,cur}}_{r,s}\right)=\mathrm{VaR}_{r,\alpha}+\frac{1}{1-\alpha}\sum_{s}\psi_{r,s}\sigma_{r,s} \tag{B.6}$$

$$\sigma_{r,s}=\max\left\{\delta^{\mathrm{rd,cur}}_{r,s}-\mathrm{VaR}_{r,\alpha},0\right\} \tag{B.7}$$

$$\begin{cases}\sigma_{r,s}\ge\delta^{\mathrm{rd,cur}}_{r,s}-\mathrm{VaR}_{r,\alpha}\\ \sigma_{r,s}\ge 0\end{cases} \tag{B.8}$$

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(a) RD coincidence factor τ_{ij}

(b) Weighted Pearson correlation coefficient ρ_{ij}

(c) Seasonal Gini coefficient ϖ_{ij}

Fig. A1. Confidence level of different indicators.

TABLE VIII
TECHNO-ECONOMIC PARAMETERS OF GENERATIONS [37]

Generation	Capex (10^4 CNY/MW)	Variable cost (10^4 CNY/MWh)	Fixed cost (10^4 CNY/(MW.yr))	Min output ratio (%)	Lifetime (yrs)
Coal	350	0.024	6.2	50%	30
Gas	200	0.045	9.8	30%	30
Hydro	1400	0	25.9	0	80
Nuclear	1750	0.005	55.6	70%	60
Utility-scale PV	420	0	6.7	0	25
Distributed PV	420	0	6.3	0	25
Onshore wind	520	0	14.6	0	25
Offshore wind	1400	0	43.5	0	25

TABLE IX
TECHNO-ECONOMIC PARAMETERS OF STORAGE

Facility	CAPEX	Annual OPEX	Lifetime (yrs)	Efficiency	Operation cycle
Battery [38]	3500 CNY/kW	0.02	15	90%	Daily
Pumped hydro [39]	5770 CNY/kW	0.02	50	75%	Daily
Hydrogen electrolyzer [40]	3000 CNY/kW	0.02	15	60%	Yearly
Hydrogen storage [41]	250 CNY/Nm^3	0.02	15	95%	Yearly

APPENDIX C: INTRA-DAY OPERATION CONSTRAINTS

Based on (25), (C.1)-(C.7) further explain the intra-day operation constraints in detail. Equation (C.1) is the power balance constraint within the typical day u. Equation (C.2) sets the upper/lower limit for the hourly load curtailment. Equation (C.3) formulates the power flow constraint. Equation (C.4) is the renewable output limit. Equation (C.5) is the unit commitment constraints within the typical day u. Equation (C.6) is the operation constraints of battery and pumped storage operating within the daily cycle. Equation (C.7) is the renewable penetration constraint. β is the renewable penetration ratio.

$$\sum_{g\in\mathcal{G}_n} p_{g,u,t} + \sum_{b\in\mathcal{B}_n} \left(p^{\text{dis}}_{b,u,t} - p^{\text{cha}}_{b,u,t}\right) +$$

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$$\left(\sum_{l\in\mathcal{L}_n^+} f_{l,u,t} - \sum_{l\in\mathcal{L}_n^-} f_{l,u,t}\right) = d_{n,u,t} - d_{n,u,t}^{\text{cur}},$$

$$g \in \mathcal{G}_n\colon \{\mathcal{T}_n,\mathcal{W}_n,\mathcal{V}_n\}, \mathcal{B}_n\colon \{\mathcal{X}_n,\mathcal{Y}_n\}, \forall n,u,t \tag{C.1}$$

$$0 \le d_{n,u,t}^{\text{cur}} \le d_{n,u,t}, \forall n,u,t \tag{C.2}$$

$$|f_{l,u,t}| \le L_l + L_l^0 \tag{C.3}$$

$$\begin{cases} 0 \le p_{g,u,t} \le G_g\omega_{r,u,t} \\ p_{g,u,t} + p_{g,u,t}^{\text{cur}} = G_g\omega_{r,u,t}^{\text{rd}}, \text{g} \in \{\mathcal{W}_r,\mathcal{V}_r\}, \forall r,u,t \end{cases} \tag{C.4}$$

$$\begin{cases} 0 \le \lambda_g o_{g,u,t} \le p_{g,u,t} \le o_{g,u,t} \le G_g \\ -\gamma_g^{\text{dn}} o_{g,u,t} \le p_{g,u,t} - p_{g,u,t-1} \le \gamma_g^{\text{up}} o_{g,u,t} \\ o_{g,u,t} = o_{g,u,t-1} + o_{g,u,t}^{\text{up}} - o_{g,u,t}^{\text{dn}} \\ o_{g,u,t} \ge \sum_{\tau=1}^{T_g^{\text{on}}} o_{g,u,t-\tau}^{\text{up}} \\ o_{g,u,t}^{\text{rd}} \le G_g - \sum_{\tau=1}^{T_g^{\text{off}}} o_{g,u,t-\tau}^{\text{dn}} \\ \forall g,u,t \end{cases} \tag{C.5}$$

$$\begin{cases} q_{b,u,t} = q_{b,u,t-1} + \eta_b^{\text{cha}} p_{b,u,t}^{\text{cha}} - p_{b,u,t}^{\text{dis}}/\eta_b^{\text{dis}} \\ 0 \le p_{b,u,t}^{\text{cha}}, p_{b,u,t}^{\text{dis}} \le B_b \\ 0 \le q_{b,u,t} \le T_b B_b \\ q_{b,u,t_{\text{start}}} = q_{b,u,t_{\text{end}}} \\ \forall u,t,b \in \mathcal{X} \end{cases} \tag{C.6}$$

$$\sum_{g,u,t} \pi_u p_{g,u,t} \ge \beta \sum_{n,u,t} \pi_u d_{n,u,t} \tag{C.7}$$

Appendix D: Boundary Information For the Optimal Planning

Table VII shows the existing generation/storage capacity within each region. Considering that the three renewable energy bases are still under construction and will be established starting from scratch, Table VII only depicts the existing capacity of the load centers in China's East Power Grid.

Tables VIII and IX depict the techno-economic parameters of the generations and storage, respectively.

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